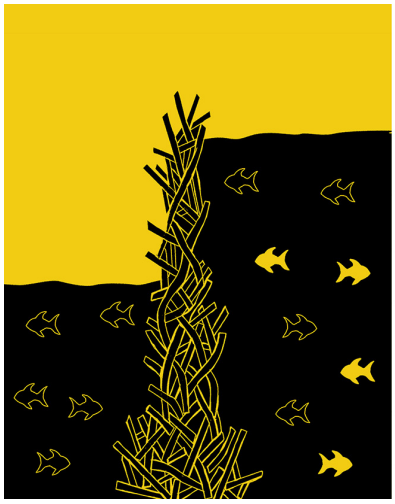


ORGANISMS CHANGE AND CONSTRUCT THE WORLD IN WHICH THEY LIVE

NICHE CONSTRUCTION

IN CONTEXT

KEY FIGURE

F. John Odling-Smee (1935–)

BEFORE

1969 British biologist Conrad Waddington writes about ways in which animals change their environments, calling this “the exploitive system.”

1983 Richard Lewontin, an American biologist, argues that organisms are active constructors of their own environments, in *Gene, Organism, and Environment*.

AFTER

2014 Canadian ecologist Blake Matthews outlines criteria for deciding whether an organism is a niche constructor.

All organisms alter the environment to cater to their own needs. Animals dig burrows, build nests, create shade from the sun, and create shelter from the wind to provide a more secure environment, while plants alter soil chemistry and cycle nutrients. When organisms modify their own and each other’s place in the environment, this is “niche construction”—a term coined by British evolutionary biologist F. John Odling-Smee in 1988.